CoursePilot: Tech Stack (Brainstorming)

1. Frontend (Dashboard)

- **Framework:** Next.js 14 (App Router) + TypeScript.
- **Styling:** Tailwind CSS + shadcn/ui (Component Library).
- **State Management:** Zustand (Local UI) + TanStack Query (Server State/Caching).
- **Key Libraries:** FullCalendar (Schedule), React Hook Form (Validation)

2. Backend (Core API)

- **Framework:** Python FastAPI (Async/Await).
- **Database:** Firebase
- **ORM:** SQLAlchemy 2.0 (Async).
- **Auth:** Firebase Auth
- **Async Tasks:** Celery + Redis (Required for non-blocking PDF processing).

3. AI & ML (AI Pipeline)

- **Orchestration:** LangChain (Python).
- **Model:** OpenAI GPT-4o-mini (Cost/Performance balance).
- **RAG/Parsing: LlamaParse** (Critical for parsing syllabus tables) + **pgvector** (Vector Search within Postgres). OR Firebase vector embeddings

4. Infrastructure & DevOps

- **Hosting:** Vercel (Frontend), Render or Railway (Backend).
- **Storage:** Supabase Storage (S3 wrapper for PDFs). OR Firebase Storage
- **CI/CD:** GitHub Actions (Linting via Ruff/ESLint; Tests via Pytest).

5. Critical Data Flow (Syllabus Ingestion)

1. **Upload:** User sends PDF -> Saved to Supabase Storage.
2. **Queue:** Backend pushes task to Redis -> Returns "202 Accepted".
3. **Process:** Celery Worker uses LlamaParse + GPT-4o to extract deadlines.
4. **Verify:** User reviews extracted dates in UI before syncing to DB.